

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses sum of probabilities = 1 to correctly find k	AO1.1b	B1	$2k + 4k + 6k + 8k = 1$ $20k = 1$
Calculates $E(Y)$ or $E(Y^2)$ correctly	AO1.1a	M1	$k = \frac{1}{20}$ or 0.05
Calculates both $E(Y)$ and $E(Y^2)$ correctly	AO1.1b	A1	$E(Y) = 1 \times 2k + 2 \times 4k + 3 \times 6k + 4 \times 8k$ $= 60k$ or 3
Calculates $\text{Var}(Y)$	AO1.1a	M1	$E(Y^2) = 1^2 \times 2k + 2^2 \times 4k + 3^2 \times 6k + 4^2 \times 8k$ $= 200k$ or 10
Uses formula for $\text{Var}(aY \pm b)$	AO1.1a	M1	$\text{Var}(Y) = E(Y^2) - E(Y)^2$ $= 200k - (60k)^2$ or $10 - 3^2$ $= 200k - 3600k^2$ or 1
Completes a fully correct, logical argument to show that $\text{Var}(5Y - 2) = 25$ AG	AO2.1	R1	$\text{Var}(5Y - 2) = 5^2 \text{Var}(Y)$ $= 5^2 \times 1$ $= 25$

ALT

Marking Instructions	AO	Marks	Typical Solution										
Uses sum of probabilities = 1 to correctly find k	AO1.1b	B1	$2k + 4k + 6k + 8k = 1$ $20k = 1$										
Finds values of $5Y - 2$ for each value of y	AO1.1a	M1	$k = \frac{1}{20}$ or 0.05 <table><tr><td>y</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$5y - 2$</td><td>3</td><td>8</td><td>13</td><td>18</td></tr></table>	y	1	2	3	4	$5y - 2$	3	8	13	18
y	1	2	3	4									
$5y - 2$	3	8	13	18									
Calculates $E(5Y - 2)$ or $E((5Y - 2)^2)$ correctly	AO1.1a	M1	$E(5Y - 2) = 3 \times 2k + 8 \times 4k + 13 \times 6k + 18 \times 8k$ $= 260k$ or 13										
Calculates $E(5Y - 2)$ and $E((5Y - 2)^2)$ correctly	AO1.1b	A1	$E((5Y - 2)^2) = 3^2 \times 2k + 8^2 \times 4k + 13^2 \times 6k + 18^2 \times 8k$ $= 3880k$ or 194										
Calculates $\text{Var}(5Y - 2)$	AO1.1a	M1	$\text{Var}(5Y - 2) = E((5Y - 2)^2) - E(5Y - 2)^2$ $= 3880k - (260k)^2$ or $194 - 13^2$ $= 3880k - 67600k^2$ $= 25$										
Completes a fully correct, logical argument to show that $\text{Var}(5Y - 2) = 25$ AG	AO2.1	R1											

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) Identifies that total probability must sum to 1	AO3.1a	M1	$c + 225k - k(15 - 12)^2 = 1$
	Substitutes $t = 12$ to form an equation in c and k .	AO1.1a	M1	$c + 216k = 1$ $1 - c = 216k$
	Rearranges equation with correct rigorous workings to show required result AG	AO2.1	R1	
(ii)	Correct expression for $E(T)$ (PI)	AO1.1a	M1	$E(T) = (-2 \times c) + \int_0^{12} t \times f(t) dt$ $f(t) = \frac{d}{dt}(225k - k(15 - t)^2)$ $= 2k(15 - t)$
	Uses $k = \frac{1}{240}$ and integrates with correct limits	AO1.1a	M1	$E(T) = -0.2 + 2k \int_0^{12} 15t - t^2 dt$ $0.9 = 216k \text{ so } k = \frac{9}{2160} = \frac{1}{240}$ $E(T) = -0.2 + \frac{2}{240} \left[\frac{15}{2} t^2 - \frac{1}{3} t^3 \right]_0^{12}$
	Obtains correct value for $E(T)$	AO1.1b	A1	$= -0.2 + \frac{1}{120} \times 504 = -0.2 + 4.2$ $E(T) = 4$

(b)	Correct expression for $E(\sqrt{ T })$	AO1.1a	M1	$E(\sqrt{ T }) = 0.1 \times \sqrt{2}$ $+ \int_0^{12} 2k \times \sqrt{t} \times (15 - t) dt$
	Integrates and uses limits	AO1.1a	M1	$= \frac{1}{120} \left(10 \times 12^{\frac{3}{2}} - \frac{2}{5} \times 12^{\frac{5}{2}} \right)$ $= \frac{12^{\frac{3}{2}}}{120} \left(10 - \frac{24}{5} \right)$
	Completes clear correct rigorous workings to show required result AG	AO2.1	R1	$= \frac{12 \times \sqrt{12}}{120} \times \frac{26}{5} = \frac{13\sqrt{12}}{25} = \frac{26\sqrt{3}}{25}$

Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.			$E(\sqrt{T}) = 0.1 \times \sqrt{2} + \frac{26\sqrt{3}}{25}$ $= \frac{\sqrt{2}}{10} + \frac{26\sqrt{3}}{25}$ $= \frac{5\sqrt{2} + 52\sqrt{3}}{50}$
Total 9 marks			

Q3.

(a) Mean = $0 \times 0.1 + 1 \times 0.35 \dots\dots$

AG

M1

= 1.85

$E(X^2) = 0^2 \times 0.1 + 1^2 \times 0.35 \dots$

Full method including $- 1.85^2$

M1

= 4.75 $\text{Var}(X) = 4.75 - 1.85^2$

For $E(X^2) = 4.75$

A1

= 1.3275

For final answer AWRT 1.33

A1

4

(b) (i) $T = c + nX$

B1

1

(ii) $E(c + nX) = c + nE(X)$

Getting at least as far as $c + E(nX)$

M1

= $c + 1.85n$

CAO

A1

$$\text{Var}(c + nX) = \text{Var}(c) + \text{Var}(nX)$$

Getting at least as far as $(0 +) \text{Var}(nx)$

M1

$$= 0 + n^2 \text{var}(X) = 1.3275n^2$$

FT their $\text{Var}(X)$ if $0 < \text{Var}(X) < 4$

A1F

4

[9]

Q4.

(a) (i) $1 - \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)$
OE

M1

$$= \frac{1}{20} = 0.05$$

AG

A1

2

(ii) $E(X) =$
 $1 \times \frac{1}{3} + 2 \times \frac{1}{4} + 3 \times \frac{1}{5} + 4 \times \frac{1}{6} + 5 \times \frac{1}{20}$
At least 2 terms

M1

$$= 2.35$$

OE: give B2 for only 2.35 seen

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A1

2

(iii) $E(X^2) =$
 $1 \times \frac{1}{3} + 4 \times \frac{1}{4} + 9 \times \frac{1}{5} + 16 \times \frac{1}{6} + 25 \times \frac{1}{20}$
(= 7.05)

All 5 terms

$E(X^2) = 7.05$ with no working scores M0

Correct working but labelled $\text{Var}(X)$ and then no more done also scores M0

M1

$$\text{Var}(X) = E(X^2) - E(X)^2$$

Applied to this problem

m1

$$= 1.5275$$

AG

A1
3

(iv) $1 - (\frac{1}{3} + \frac{1}{4})$ or $(\frac{1}{5} + \frac{1}{6} + \frac{1}{20})$

M1

$= \frac{5}{12}$ or 0.417

AWRT Accept answer only for B2

A1
2

(b) $'2.35' \times 100 - 50$

Their value of mean

M1

$= 185$

FT from (a)(ii)
Give B2 for only 185 seen

A1F

$100^2 \times 1.5275$ or $100 \times \sqrt{1.5275}$

M1

$SD = \sqrt{15275} = 5\sqrt{611} = 124$

AWFW 123.5 to 124 or $5\sqrt{611}$

Give B2 for only 123.5 to 124 or $5\sqrt{611}$ seen

A1
4

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[13]

Q5.

(a) (i) Poisson

B1
1

(ii) $E(3X - 1) = 3\lambda - 1$

B1

$Var(3X - 1) = 9\lambda$

oe (allow $3^2\lambda$)

B1
2

$$(iii) \quad P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$$

B1

$$\begin{aligned} P(X = x + 1) &= \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!} \\ &= \frac{e^{-\lambda} \times \lambda^x \times \lambda}{(x+1)x!} \\ &= \frac{\lambda}{x+1} \times \frac{e^{-\lambda} \times \lambda^x}{x!} \end{aligned}$$

dep(B1)

Mdep1

$$= \frac{\lambda}{x+1} P(X = x)$$

AG

Adep1

3

(b) (i) $\lambda_{\text{car}} = 500 / \text{hour}$

$$\lambda_{\text{coach}} = 10 / \text{hour}$$

$$\Rightarrow \lambda_{\text{vehicle}} = 510 / \text{hour} = 8.5 / \text{min}$$

for 8.5 stated / used

B1

$$P(V \geq 10) = 1 - 0.6530$$

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M1

$$= 0.347$$

A1

special case:

$$\lambda = 10 \Rightarrow B1M0A0$$

$$B1 \Rightarrow 1 - 0.458 \text{ or } 0.542$$

3

(ii) $\mu_{\text{car}} = 836 / \text{hour}$

$$\mu_{\text{coach}} = 22 / \text{hour}$$

$$\Rightarrow \mu_{\text{vehicle}} = 858 / \text{hour} = 14.3 / \text{min}$$

for 14.3 stated / used

B1

$$P(V \leq 3) = P(V = 0, 1, 2, 3)$$

$$= \begin{cases} e^{-14.3} \left[1 + \frac{14.3}{1} + \frac{14.3^2}{2} + \frac{14.3^3}{6} \right] \\ e^{-14.3} \times 604.91283 \end{cases}$$

$$= 0.0003726 \text{ to } 0.000373$$

All 4 terms required for **any** $\lambda > 0$
M0 for use of normal approximation

M1

$$= 0.00037 \text{ (2sf)}$$

dep M1

Adep1

3

[12]

Q6.

(a)

<i>r</i>	1	2	3	4	5
<i>p</i>	.5	.24	.144	.0864	0.0296

$$0.4 \times 0.6 = 0.24$$

$$0.24 \times 0.6 = 0.144$$

$$0.144 \times 0.6 = 0.0864$$

B1 for any 1 correct (unsimplified) (B1)

B2 all correct and simplified

B2,1

2

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(b) P(fewer than 3 bedrooms **are** rented)

$$= P(R = 1, 2) \Rightarrow$$

P(fewer than 3 bedrooms **not** rented)

$$= 1 - P(R = 1, 2)$$

M1

$$= 1 - P(1 \text{ or } 2 \text{ rooms are rented})$$

$$= 1 - (0.5 + \mathbf{0.24})$$

[their $0 < p(2) \leq 0.4704$ value from table **used**]

$$= 1 - 0.74$$

m1

$$= 0.26$$

A1

Alternative:

$P(\text{fewer than 3 not rented})$

$$= P(0, 1 \text{ or } 2 \text{ not rented})$$

$$= P(5, 4 \text{ or } 3 \text{ are rented})$$

$$= P(R = 3, 4, 5) \quad \text{M1}$$

$$p = 0.4 \times 0.6^2 + 0.4 \times 0.6^3 + 0.0296$$

$$= 0.144 + 0.0864 + 0.0296 \quad \text{m1}$$

[or their $p(3) + p(4) \leq 0.4704$ value from table used]

$$= 0.26 \text{ (cao)} \quad \text{A1}$$

[SC 0.74 for B1]

3

(c) (i) $E(R) = 0.5 \times 1 + 0.4 \times 0.6 \times 2 + 0.4 \times 0.6^2 \times 3 + 0.4 \times 0.6^3 \times 4 + 0.0296 \times 5$
 $= 0.5 \times 1 + 0.24 \times 2 + 0.144 \times 3 + 0.0864 \times 4 + 0.0296 \times 5$

$$\sum_{i=1}^5 r_i \times P(R = r_i) \text{ from their table}$$

M1

$$= 0.5 + 0.48 + 0.432 + 0.3456 + 0.148$$

$$\left[= \frac{1}{2} + \frac{12}{25} + \frac{54}{125} + \frac{216}{625} + \frac{37}{250} \right]$$

$$(0.5 + 1.2576 + 0.148)$$

$$\therefore E(R) = 1.9056$$

$$[\text{awfw } 1.9 \text{ to } 1.91] \left[1 \frac{566}{625} \right]$$

A1

2

(ii) $E(R^2) = 0.5 \times 1^2 + 0.4 \times 0.6 \times 2^2 + 0.4 \times 0.6^2 \times 3^2 + 0.4 \times 0.6^3 \times 4^2 + 0.0296 \times 5^2$
 $[0.5 + 0.96 + 1.296 + 1.3824 + 0.74]$

$$E(R^2) = 4.8784$$

AG

B1

$$\text{Var}(R) = 4.8784 - 1.9056^2 (= 1.24708864)$$

$$4.8784 - \text{their } E^2(R)$$

M1

$$= 1.25(3\text{sf})$$

$$(\text{awfw } 1.23 \text{ to } 1.25)$$

A1

3

(d) $E(M) = 1250E(R) - 282 = 1250 \times 1.9056 - 282 = 2100$

cao

B1

$$\begin{aligned}\text{Var}(M) &= 1250^2 \times [4.8784 - 1.9056^2] \\ 1250^2 \times \text{their Var}(R) &> 0 \text{ in (c)(ii)} \\ (1\,948\,473 \text{ to } 1\,953\,125)\end{aligned}$$

$$\begin{aligned}\text{sd}(M) &= 1250 \times \sqrt{1.24708864} \\ \text{sd}(M) &= \sqrt{1948437} = 1395.9 \\ (\sqrt{1953125} &= 1397.5)\end{aligned}$$

$$\begin{aligned}&= 1395.91 \\ &(\text{awfw } 1395 \text{ to } 1400)\end{aligned}$$

A1

3

[13]

Q7.

(a)

r	1	2	3	4
$P(R=r)$	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$	k

$$k + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} = 1 \Rightarrow k = \frac{1}{27}$$

M1

AG

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A1

2

$$\begin{aligned}\text{(b)} \quad P(R \geq 3) &= \frac{2}{27} + \frac{1}{27} = \frac{1}{9} \\ &\frac{3}{27} \\ &\text{Allow } \frac{3}{27} \text{ or } 0.111\end{aligned}$$

B1

1

$$\text{(c) (i) } C = 27R + 5$$

$$\begin{aligned}E(R) &= \left(1 \times \frac{2}{3}\right) + \left(2 \times \frac{2}{9}\right) + \left(3 \times \frac{2}{27}\right) + \left(4 \times \frac{1}{27}\right) \\ &= 1\frac{13}{27}\end{aligned}$$

$$(1.48) \text{ or } \frac{40}{27}$$

B1

$$\therefore E(C) = 27 \times 1 \frac{13}{27} + 5$$

M1

$$= 45$$

A1F

3

$$\begin{aligned}
 \text{(ii)} \quad E(R^2) &= \left(1 \times \frac{2}{3}\right) + \left(4 \times \frac{2}{9}\right) + \left(9 \times \frac{2}{27}\right) + \left(16 \times \frac{1}{27}\right) \\
 &= 2 \frac{22}{27} \text{ or } \frac{76}{27} \\
 &\quad (2.81)
 \end{aligned}$$

B1

$$\text{Var}(R) = 2 \frac{22}{27} - \left(1 \frac{13}{27}\right)^2$$

M1

$$= \frac{452}{729}$$

(0.62)

$$\therefore \text{St. dev}^n(C) = 27 \times \sqrt{\frac{452}{729}}$$

$$27 \times \sqrt{\text{Var}(R)} [\text{Var}(R) > 0]$$

M1

$$= 21.3$$

CAO (21.26)

A1

4

$$\begin{aligned}
 \text{SC: } \text{Var}(C) &= 452 \text{ (CAO)} \\
 &\text{(B1M1B1A0)}
 \end{aligned}$$

Alternative (c)

C	32	59	86	113
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P	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$	$\frac{1}{27}$
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C	32	59	86	113
n	18	6	2	1

$$\bar{x} = 45 \text{ and } \sigma = 21.260$$

from calculator

$$\left(\bar{x} = \frac{\sum Cn}{27} \right)$$

[10]

Q8.

(a) (i) $P(\text{GG or YY or RR})$

$$= \frac{2}{10} \times \frac{1}{9} + \frac{3}{10} \times \frac{2}{9} + \frac{4}{10} \times \frac{3}{9}$$

$$= \frac{2}{9}$$

(AG)

M1

A1

2

$$(ii) \quad P(\bar{B}\bar{B} \text{ or } \bar{B}B) = \frac{1}{10} \times \frac{9}{9} + \frac{9}{10} \times \frac{1}{9}$$

$$= \frac{1}{10} + \frac{9}{10} \times \frac{1}{9}$$

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M1

$$= \frac{1}{5}$$

(AG)

A1

2

(b) (i)

	Same	1 Blue	Neither
x	135	145	-45
$P(X = x)$	$\frac{2}{9}$	$\frac{1}{5}$	$\frac{26}{45}$

B1B1

2

$$(ii) \quad E(X) = 135 \times \frac{2}{9} + 145 \times \frac{1}{5} + (-45) \times \frac{26}{45}$$

Multiply two rows of their table from (b)(i)

M1

$$= 29 + 30 - 26 = 33 \text{ pence}$$

AG

A1

2

(c) (i) $E(Y) = 104 - 3E(X)$

M1

$$= 104 - 3 \times 33 = 5 \text{ pence}$$

A1

$\therefore$ Joanne would expect to win £5

OE (eg 500p)

A1

3

(ii) $E(X^2) = 9425$
 $(4205 + 4050 + 1170)$

B1

$$\text{Var}(X) = 9425 - 33^2 = \mathbf{8336}$$

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$$sd(X) = 91.30$$

$$\text{Var}(Y) = 9 \times \text{Var}(X)$$

B1

$$= 9 \times 8336$$

$$9 \times (\text{their } \text{Var}(X) > 0)$$

$$= 75024$$

$$\text{or } 3 \times (\text{their } sd(X))$$

M1

$$\Rightarrow \text{standard deviation } (Y) = 274 \text{ pence}$$

$$273.9p \text{ or } £2.74$$

A1

4

[15]

Q9.

(a) (i) $E(Y) = \sum y P(Y = y)$
 $= 5 \times 0.1 + 15 \times 0.2 + 25 \times 0.3 + 35 \times 0.4$
 $= 25$

B1

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2$$

$$= 725 - 25^2$$

M1

$$= 100$$

CAO

A1

Standard deviation = 10

ft on $\text{Var}(Y) > 0$

A1ft

(ii) $C = 10Y + 5$
 $E(C) = 10E(Y) + 5$
 $= 10 \times 25 + 5$
 $= 255 \text{ pence}$

OE

B1

1

(b) $\text{Var}(X) = E(X^2) - [E(X)]^2$
 $= 75.25 - 8.35^2$

M1

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$$= 75.25 - 69.7225$$

$$= 5.5275$$

AWFW 5.52 to 5.53

A1

$$T = 0.4X + 250$$

$$\text{Var}(T) = \text{Var}(0.4X + 250)$$

$$= 0.4^2 \times \text{Var}(X)$$

$\text{Var}(X) > 0$

M1

$$= 0.16 \times 5.5275$$

$$= 0.8844$$

AWFW 0.884 to 0.885

A1

4

Q10.

(a) $\sum p = 1$
 $k = 1 - (0.40 + 0.25 + 0.18 + 0.12)$
 $k = 0.05$

B1

1

(b) (i) $E(X) = \sum_{\text{all } x} x P(X = x) = 3.17$

B1

1

(ii)
$$\text{Var}(X) = \sum_{\text{all } x} x^2 P(X = x) - \mu^2$$

$$\sum x^2 P(X = x) \text{ attempted}$$

M1

$$= 11.53 - 10.0489$$

$$\sum x^2 P(X = x) - [E(X)]^2$$

M1

$$= 1.4811$$

$$= 1.48$$

$$\text{AWFW } 1.48 - 1.49$$

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A1

3

(c) (i) $E(Y) = 2E(X) - 3$
 $= 2 \times 3.17 - 3$
 $= 3.34$

M1

1

(ii) $\text{Var}(Y) = 4 \text{Var}(X)$
their $\text{Var}(X) \times 4$

M1

$$= 4 \times 1.48$$

$$= 5.92$$

$$\text{Standard deviation} = 2.43$$

$$\sqrt{\quad}$$

Note: if separate table used for Y:

$$E(Y^2) = 17.08$$

$$[E(Y)]^2 = 11.16$$

m1A1

3

[9]

Q11.

- (a) $\sum p = 1$
 $k = 1 - (0.01 + 0.05 + 0.14 + 0.30 + 0.12)$
 $k = 0.38$

B1

1

(b) (i) $E(X) = \sum_{\text{all } x} x P(X = x) = 4.35$
 $\frac{87}{20}$

B1

1

(ii) $\text{Var}(X) = \sum_{\text{all } x} x^2 P(X = x) - \mu^2$
 $E(X^2) \text{ attempted}$

M1

$$= 20.09 - 18.9225$$

$$\sum x^2 P(X = x) - \mu^2$$

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M1

$$= 1.1675$$

$$\frac{467}{400} (\text{AWFW } 1.16 - 1.17)$$

A1

3

- (c) (i) $E(Y) = 5E(X) + 2$
 $= 5 \times 4.35 + 2$
 $= 23.75$
Their (b)(i) $\times 5 + 2$

M1

1

(ii) $\text{Var}(Y) = 25\text{Var}(X)$
 $= 29.1875$

Their (b)(ii) $\times 25$

M1

Standard deviation = 5.40

$\sqrt{\quad}$

m1

(5.40 – 5.41)

A1

3

[9]

Q12.

(a) $E(X) = \sum_{\text{all } x} x \cdot P(X = x)$
= 50

CAO

B1

$$E(X^2) = \sum_{\text{all } x} x^2 \cdot P(X = x)$$

$$= 2602.6(0)$$

M1

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 2602.6 - 50^2$$

$$= 102.6(0)$$

M1

$\Rightarrow$ standard deviation (X) = 10.13
(to nearest 1p)

A1

4

(b) $E(Y) = \mu = E(10X + 250)$
= $10 \times E(X) + 250$
= 750
On their $E(X)$

B1ft

$$\text{s.d.}(Y) = 10 \times 10.1$$

= 101

On their $sd(X)$

B1ft

2

[6]

Q13.

(a) $L \sim (580, 1600) \quad G = \frac{L}{2} + 80$

$$E(G) = \frac{580}{2} + 80 = 370$$

CAO

B1

$$V(G) = \frac{V(L)}{4}$$

use of $V(aX + b) = a^2 \times V(X)$ with $a \neq 1$ and $b > 0$

$$= 400$$

M1

A1

3

(b) (i) $R = 1000 - L - G$

use of

$$= 920 - \frac{3L}{2}$$

CAO; OE

M1

A1

2

(ii) $E(R) = 920 - \frac{3 \times 580}{2} = 50$

CAO

B1

$$V(R) = \frac{9 \times V(L)}{4}$$

Use of $V(aX + b) = a^2 \times V(X)$ with
 $a \neq \pm 1$ and $b > 0$
applied to expression for R

M1

$$= 3600$$

CAO

Note:

$$\sqrt{1600} + \sqrt{400} = 40 + 20 = 60$$

$$\text{and } 60^2 = 3600$$

A1

3

[8]

Q14.

(a)	$r:$	0	1	2	3	4
	$P(R = r)$	0.1	0.2	0.4	0.2	0.1

(i) $E(R) = 0 + 0.2 + 0.8 + 0.6 + 0.4 = 2$

AG; use of $\sum r \times p_r$ or symmetrical argument

M1

1

(ii) $E(R^2) = 0 + 0.2 + 1.6 + 1.8 + 1.6 = 5.2$

CAO; must be some evidence of use of $\sum r^2 \times p_r$

B1

$$\text{Var}(R) = E(R^2) - (E(R))^2 = 1.2$$

AG; use of a formula for $\text{Var}(R)$

M1

or

$$= 0.4 + 0.2 + 0 + 0.2 + 0.4 = 1.2$$

CAO; ≥ 4 terms correct

(B1)

2

(b) $E(P) = 3 \times 2 + 4 = 10$

CAO

B1

$$\text{Var}(P) = 3^2 \times \text{Var}(R)$$

Use of $\text{Var}(aX + b) = a^2 \text{Var}(X)$

with $a > 1$ and $b \geq 0$

M1

$$= 10.8$$

CAO

A1

- (c) (i) $C = 200 - R - P$
Use of; may be implied

M1

$$= 200 - R - (3R + 4)$$

Hence $C = 196 - 4R$
CAO

A1

2

- (ii) $E(C) = 196 - 4 \times 2 = 188$
CAO

B1

$$\text{Var}(C) = 4^2 \times \text{Var}(R) = 19.2$$

CAO; dependent on A1 in (c)(i)

B1dep

2

[10]

Q15.

- (a) Profit = Sales – Cost + Refund
Use of at least (Sales – Cost)

M1

$$\therefore P = 2.20S - (20 \times 1.00)$$

$2.2S - 20$; or equivalent

A1

$$+ (20 - S) \times 0.20$$

$4 - 0.2S$; or equivalent

A1

3

$$= 2S - 16$$

AG

- (b) $E(P) = 2 \times E(S) - 16 = £14.00$
CAO, accept 14 (ignore units)

B1

$$\text{Var}(P) = 2^2 \times \text{Var}(S)$$

$\text{Var}(aX - b) = a^2 \times \text{Var}(X)$ with $a > 1$

M1

$$= £^216.00$$

CAO; accept 16 (ignore units)

A1

$$SD(P) = \sqrt{\text{Var}(P)} = \sqrt{16} = £4.00$$

Use of $\sqrt{\text{Var}(P)}$

m1

4

[7]

Q16.

(a)

$v:$	1	2	3	4	5
$p:$	0.15	0.45	0.20	0.15	0.05

$$E(V) = \sum vp = 2.5$$

Evidence of $\sum vp$ used; AG

M1

$$E(V^2) = \sum v^2p = 7.4$$

CAO or implied correctly

B1

$$\therefore \text{Var}(V) = E(V^2) - (E(V))^2$$

Evidence of $E(V^2) - (E(V))^2$ used

M1

3

$$= 7.4 - 2.5^2 = 1.15$$

AG

(b) $E(C) = (30 \times 2.5) + 25 = 100$

CAO

B1

$$\text{Var}(C) = 30^2 \text{Var}(V)$$

$$= \text{Var}(aX + b) = a^2 \text{Var}(X)$$

M1

$$= 1035$$

CAO

A1

3

(c) $E(R) = 40E(V^2) + 15E(V) + 50$

use of, OE

M1

$$= (40 \times 7.4) + (15 \times 2.5) + 50$$

f.t. on $E(V^2)$, or B1 for 105, 240, 455, 750, 1125

A1f.t.

$$= 383.5$$

CAO
NB ignore units in (b) & (c)

A1

3

[9]

Q17.

(a) (i) $E(M_1 + M_2 + M_3) = 3 \times 1.7 = 5.1$

B1

1

(ii) $\text{Var}(M_1 + M_2 + M_3) = 3 \times 0.81 = 2.43$

B1

1

(b) (i) $E(R) = \sum p \times R$
 $= 2 \times \frac{1}{8} + 4 \times \frac{9}{16} + 6 \times \frac{5}{16}$

M1

$$= 4\frac{3}{8}$$

4.375 (AWRT 4.38)

A1

$$\text{Var}(R) = \sum p \times R^2 - E^2(R)$$

$$= 4 \times \frac{1}{8} + 16 \times \frac{9}{16} + 36 \times \frac{5}{16} - \left(4\frac{3}{8}\right)^2$$

M1

$$= 20\frac{3}{4} - 19\frac{9}{64}$$

$$= 1\frac{39}{64}$$

1.609375 (AWRT 1.61)

A1

4

$$(ii) \quad E\left(\frac{1}{2}R\right) = \frac{1}{2} E(R) = \frac{1}{2} \times 4\frac{3}{8} = 2\frac{3}{16}$$

B1f.t.

$$\text{Var}\left(\frac{1}{2}R\right) = \frac{1}{4} \times \text{Var}(R)$$

$$= \frac{1}{4} \times 1\frac{39}{64}$$

M1

$$= \frac{103}{256}$$

0.40234 (AWRT 0.402)
(on their (b)(i) Var(R) value)

A1f.t.

3

$$(c) \quad (i) \quad T = \frac{1}{2}R + M_1 + M_2 + M_3$$

$$E(T) = E\left(\frac{1}{2}R\right) + E(M_1 + M_2 + M_3)$$

$$= 2\frac{3}{16} + 5.1$$

$$= 7.29$$

(AWRT 7.29)

7.2875 (a)(i) + (b)(ii)

$$(if (b)(ii) \quad E\left(\frac{1}{2}R\right) = \frac{1}{2} (E(R) \text{ used.}))$$

B1f.t.

1

$$(ii) \quad \text{Var}(T) = \text{Var}\left(\frac{1}{2}R\right) + \text{Var}(M_1 + M_2 + M_3)$$

$$= \frac{103}{256} + 2.43$$

(their a(ii) + b(ii))

M1

$$= 2.832$$

(AWRT 2.83)

Q18.

- (a) $E(R) = 3$ $\text{Var}(R) = 1.2$
 $E(5(2R - 1)) = E(10R - 5) = 25$
 CAO

B1

$\text{Var}(10R - 5) = 10^2 \text{Var}(R)$
 use of $\text{Var}(aX \pm b) = a^2 \text{Var}(X)$ with $a > 1$ and $b \geq 0$

M1

$= 120$

CAO

	60	30	20	15	2
$\frac{60}{r}$					
(b) (i)	60	30	20	15	2
p :	0.1	0.2	0.4	0.2	0.1

CAO; OE; must be stated*
 CAO; must be stated; dep on previous B1*
 (*)not from (ii) unless tables

A1

3

B1

B1

2

- (ii) $E\left(\frac{60}{R}\right) = \sum \frac{60}{r} \times p$
 use of values of p must be correct

M1

1

$= 24.2$
 AG

- (iii) $\text{Var}\left(\frac{60}{R}\right) = E\left(\frac{3600}{R^2}\right) - \left(\frac{60}{R}\right)^2$
 $= 759.4 - 24.2^2$
 use of

M1

= 173.7 to 173.8

AWFW

exact answer exact answer is 173.76

do not accept 173 or 174

A1

2

[8]

Q19.

- (a) All digits are equally likely
So
by symmetry

$$P(D = d) = \frac{1}{9}$$

So

$$E(D) = \frac{45}{9} = 5$$

B1

B1

2

- (b) $\text{Var}(D) = E(D^2) - (E(D))^2$
Use of

M1

$$= \frac{95}{3} - 5^2 = \frac{20}{3} \text{ or } 6\frac{2}{3}$$

CAO

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Decimal must clearly indicate $\frac{2}{3}$

A1

2

- (c) $N = 3(D + 5)$
 $E(N) = 3 \times 5 + 15 = 30$
CAO

B1

$$\text{Var}(N) = 3^2 \times \text{Var}(D)$$

Use of $\text{Var}(aX + b) = a^2\text{Var}(X)$
with $a > 1$ and $b \geq 0$

M1

$$= 9 \times \frac{20}{3} = 60$$

CAO

A1

3

[7]

Q20.

(a) $E(R) = 3$ $\text{Var}(R) = 1$

$$E(5(R - 1)) = \underline{10}$$

CAO

B1

$$\text{Var}(5(R - 1)) = 5^2 \times \text{Var}(R)$$

use of, with $a > 1$,
 $\text{Var}(aX - b) = a^2 \text{Var}(X)$

$$= 25$$

CAO

M1

A1

3

(b) $f(r) = r / 10, r = 1, 2, 3, 4$

$$E(12R^{-1}) = 12 \sum \frac{P(R=r)}{r} =$$

use of $\sum(f(r)/r)$
 (M0 for $12/3 = 4$)

M1

$$12 \times \left(\frac{1}{10} + \frac{2}{20} + \frac{3}{30} + \frac{4}{40} = \underline{4.8} \right)$$

CAO

$$\frac{4+1}{2} = 2.5 \quad \frac{12}{2.5} = 4.8 \text{ gets M0A0}$$

A1

$$E(12R^{-1})^2 = 144 \sum \frac{P(R=r)}{r^2}$$

use of $\sum(f(r)/r^2)$
 (M0 for $144/1$ or $12/1$ or $144/10$)

M1

$$= 144 \left(\frac{1}{10} + \frac{2}{40} + \frac{3}{90} + \frac{4}{160} \right) = 30$$

$$\text{Var}(12R^{-1}) =$$

$$= E(12R^{-1})^2 - (E(12R^{-1}))^2 =$$

use of

m1

$$30 - 4.8^2 = 6.96$$

CAO

A1

5

[8]

Q21.

(a) $E(T) = 5$ $\text{Var}(T) = 25$

$$P = 4T + (2T + 10) = 6T + 10$$

CAO

B1

Thus $E(P) = 6E(T) + 10 = 40$

CAO

B1

and $\text{Var}(P) = 6^2 \text{Var}(T)$

*Use of $(aX + b) = a^2 \text{Var}(X)$
with $a > 1$*

M1

$$= 36 \times 25 = 900$$

CAO

A1

4

(b) (i) $\text{Var}(T) = E(T^2) - (E(T))^2$

Use of; OE

M1

$$\text{Thus } E(T^2) = 25 + 5^2 + 50$$

CAO; AG

A1

2

(ii) $A = 2T \times (T + 5) = 2T^2 + 10T$

CAO

B1

Thus $E(A) = 2E(T^2) + 10E(T)$

Use of E on $(aT^2 + bT)$

M1

$$= 100 + 50 = 150$$

CAO

A1

3

[9]



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Examiner reports

Q1.

The question was attempted well by most students with many scoring full marks. Some students lost the final mark by writing the correct calculations but not making clear what the calculations were finding. Other students lost the final mark by writing X when referring to the random variable Y . A minority of students mistook the random variable as continuous when calculating expectation and variance whilst treating it as discrete when finding k . Some students simply added the probabilities when attempting to find the expectation and found the expectation when attempting to find the variance. Very few students attempted the question by finding the values of $5Y - 2$ first and most of these were unsuccessful.

Q3.

Part (a) was almost universally answered correctly. Most students then went on to complete the question successfully, although some found the algebra required confusing and squared the wrong part of the variance. Others ignored the instruction in part (b)(ii) to use the results from part (a) and so inevitably lost marks.

Q4.

The majority of students scored very well on this question, although a few treated X as a continuous variable and performed various integrations and so scored few, if any, marks. Students should have appreciated that, in part (a)(iii), where they were required to 'Show that $\text{Var}(X) = 1.5275$ ', simply quoting values such as $E(X^2) = 7.05$ without any working, was not acceptable. This 'general principal' can be expected to apply on future papers. Part (b) was well done, although a small proportion of students, having found the variance, did not proceed to the standard deviation.

Q5.

In part (a)(i), the vast majority of candidates correctly stated the distribution of X to be 'Poisson'. In part (a)(ii), although most stated the answers correctly as $E(3X - 1) = 3\lambda$ and $\text{Var}(3X - 1) = 9\lambda$ some, who seemed to understand the underlying concepts by stating $E(3X - 1) = 3 \times E(X) - 1$ and $\text{Var}(3X - 1) = 3^2 \times \text{Var}(X)$, then failed to state their answers in terms of λ as requested, consequently gaining no marks. Part (a)(iii) proved to be very

popular with most candidates. They were able to correctly state $P(X = x + 1) = \frac{e^{-\lambda} \times \lambda^{x+1}}{(x+1)!}$ and then were able to use

$\lambda^{x+1} = \lambda \times \lambda^x$ and $(x + 1)! = (x + 1) \times x!$ to prove the given answer.

In part (b), it was very pleasing to see that candidates dealt with such phrases as 'at least 10' and 'at most 3' much better than in previous series. In part (b)(i), $\lambda = 8.5$ followed by a correct calculation was often seen. In part (b)(ii), $\lambda = 14.3$ was correctly seen and often used to find the correct answer of 0.00037. Unfortunately, some failed to comply with the instruction 'give your answer to two significant figures' and consequently lost a mark. Those who chose to do the whole of this part of the question on their calculators either gained full marks when the correct answer was stated or lost at least two marks for

incorrect answers with no method shown.

Q6.

This was the best answered question on the paper with many fully correct solutions seen. Almost all students managed to complete the table in part (a) correctly. Weaker students found part (b) very difficult to fathom. However, there were very many fully correct solutions with most making good use of the probabilities found in part (a). Part (c) was especially well done with many correct solutions to all parts. However, some students lost credit for using premature approximation or for failing to write down the value of the standard deviation having correctly found the value for the variance.

Q7.

Far too many candidates treated the given distribution as *continuous* rather than *discrete* and consequently attempted to apply integration methods throughout. This obviously met with no success but also wasted a lot of time. Part (a) was usually done quite well. However, it must be pointed out that when an answer is given in the question, then an appropriate method must be shown by candidates in order to gain full credit. Simply evaluating $P(R = r)$ for $r = 1, 2, 3$ and then writing down the given answer was not

sufficient; some mention of $\sum_{\text{all } r} P(R = r) = 1$ had to be seen. $P(R = 4) = \frac{2}{3} \left(\frac{1}{3}\right)^{4-1} = \frac{1}{27} = k$ was also seen but again gained no marks.

In part (b), some candidates incorrectly interpreted 'at least 3' as either meaning 'greater than 3' or 'less than or equal to 3' or even 'equal to 3', with $\frac{1}{27}$, $\frac{1}{26}$ and $\frac{2}{27}$ as common wrong answers.

There were very many correct solutions seen to parts (c)(i) and (c)(ii). The most common cause for loss of marks was in part (c)(ii), where a number of candidates correctly found $\text{Var}(C) = 452$ but then did not go on to attempt to find the standard deviation of C as requested. Several candidates confused the use of R and C .

Q8.

This was often the worst answered question on the paper. Many candidates were unable to show enough convincing working in part (a) to gain full credit. However, part (b) was usually done quite well, except by those candidates who simply wrote down a 2×2 table or stated +45 rather than -45. Most realised that the probabilities had to add up to one, so correctly obtaining $\frac{26}{45}$.

The values in the table were then usually used correctly to obtain the given value of $E(X)$. Part (c)(i) was usually done quite well with many finding $E(Y) = 5$. Unfortunately some candidates failed to multiply this answer by 100 to give the final answer of 500 pence or £5. There were some excellent solutions, by various methods, to the final part of this question, but these were mainly given by the more able candidates.

Q9.

This question proved to be a good source of marks for most candidates. In part (a)(i), the

great majority gained full marks. However, there was a minority who, having found correctly $\text{Var}(Y) = 100$, failed to identify the standard deviation. Part (a)(ii) was done well by most candidates with $E(C) = 255$ often seen. In part (b), the majority of candidates found the value for $\text{Var}(X)$ and then went on to use $\text{Var}(T) = 0.4^2 \times \text{Var}(X)$ to gain the correct answer of 0.884. Those candidates who attempted to use $\text{Var}(T) = E(T^2) - [E(T)]^2$, a valid if not the most efficient method, correctly found $E(T) = 253.34$ but were then unable to show that $E(T^2) = E(0.16X^2 + 200X + 62500) = 64182.04$. As a result they often ended up with a negative answer for the variance.

Q10.

Candidates found this question very much to their liking, with the vast majority gaining full marks. Most candidates easily obtained 0.05 for the value of k in part (a), before using this value to help to obtain the correct answer of 3.17 for $E(X)$ in part (b)(i). Almost all

candidates managed to calculate $E(X^2) = \sum x^2 \times P(X = x)$ in part (b)(ii), with very few wrongly thinking that this was the same as the variance. In part (c), although the majority of candidates used the most efficient ways of finding $E(Y)$ and $\text{SD}(Y)$, by evaluating $E(Y) = 2 \times E(X) - 3$ and $\text{SD}(Y) = \sqrt{4 \times \text{Var}(X)} = 2\sqrt{\text{Var}(X)}$, there were frequent attempts made to tabulate the probability distribution for Y followed by the use of methods already employed in part (b). Although this process was often successful, it used up valuable time which could have been allocated to other more demanding parts of the paper.

Q11.

This proved to be a very popular question, with the overwhelming majority of candidates gaining full marks. Parts (a) and (b)(i) were done well, with very few numerical errors

seen. Almost all candidates managed to calculate $E(X^2) = \sum x^2 \times P(X = x)$ in part (b)(ii), with very few candidates wrongly thinking that this was the same as the variance.

In part (c), although the majority of candidates used the most efficient ways of finding $E(Y)$ and $\text{SD}(Y)$ by evaluating $E(Y) = 5 \times E(X) + 2$ and $\text{SD}(Y) = \sqrt{25 \times \text{Var}(X)} = 5 \times \sqrt{\text{Var}(X)}$, there were frequent attempts made to tabulate the probability distribution for Y , followed by the use of methods already employed in part (b). Although this process was often successful, it used up valuable time which could have been allocated to other more demanding parts of the paper. A few confused candidates thought that $\text{Var}(Y) = 25 \times E(X)$, whilst others simply found the variance, not realising that the stand requested.

Q12.

This was the best answered question on the paper. In part (a), candidates usually found $E(X) = 50$, with most going on to show the correct value of the standard deviation to be 10.13. In part (b), many correct answers were seen, with most candidates using their knowledge of the mean and variance of $(aX + b)$ to show that $E(Y) = 750$ and that the required value of the standard deviation was 101.3. A minority of candidates constructed a new table for the distribution of Y , and then repeated methods used in part (a), often with considerable or complete success.

Q13.

This proved to be the least well answered question on the paper. Whilst many candidates scored one mark in part (a) for $E(G) = 370$, most then found an incorrect value, such as

800 or 880, for $\text{Var}(G)$. In part (b)(i) it appeared that many candidates could not interpret the request correctly. Those who began correctly with $1000 - L - G$ then often displayed

$$G = \frac{L}{2} + 80$$

poor algebraic skills in the substitution of $G = \frac{L}{2} + 80$ and subsequent simplification. This, together with another application of an incorrect methodology for the variance of an expression, resulted in few candidates scoring any of the available three marks.

Q14.

Most candidates scored the first six marks for correct answers to parts (a) and (b). However, some incorrect methods occurred in part (a) with the calculation of $\text{Var}(R)$ as

$\sum (r \times P(R=r))^2$ which, perchance, equalled 1.2. Correct attempts at part (c) were much rarer and generally only the more able candidates achieved significant success. In part (i), simple algebraic manipulation proved to be a major stumbling block. In part (ii), some candidates obtained 188 for $E(C)$ directly from parts (a)(i) and (b). However, the mark for $\text{Var}(C)$ was only available to the small proportion of the candidates who scored two marks in part (i). Some of these candidates then failed to obtain 19.2, apparently thrown by $(-4R)$.

Q15.

Many candidates could not deduce that Profit = Sales – Cost + Refund, or equivalent, and so made little, if any, progress in part (a). Answers to part (b) however usually scored at least three of the four marks available. The lost mark was generally for quoting the variance as (£)16, rather than a standard deviation of £4. However, here as elsewhere on the paper, units were not required for full marks.

Q16.

Most candidates scored the 3 marks in part (a) without resorting to fiddling answers, but a small minority expressed $E(V^2)$ as $\sum v \times (P(V=v))^2$.

In part (b), whilst most candidates found $E(C)$ correctly, many candidates lost the 2 marks for $\text{Var}(C)$ by either using 30 rather than 30^2 or by adding 25.

Part (c) proved much more difficult, with approximately half the candidates scoring no marks. By far the most common error was to substitute 2.5^2 for $E(V^2)$ rather than 7.4 as found in part (a). A small number of candidates chose to set up the probability distribution for R and then find $E(R)$ from first principles. Few candidates had complete success using this valid alternative approach.

Q17.

The answers seen to this question were much improved on those seen to similar questions on previous papers.

In part (a)(i) most candidates gained the correct answer of 5.1. Although there were many correct answers of 2.43 seen in answers to part (a)(ii), there was a considerable number of candidates who thought that the required variance could be found by using $\text{Var}(M_1 + M_2 + M_3) = 3^2 \times \text{Var}(M)$.

Most candidates calculated $E(R)$ and $\text{Var}(R)$ correctly with only a few thinking that

$\text{Var}(R) = E(R^2)$. There were many correct solutions to part (b)(ii), with $E\left(\frac{1}{2}R\right) = \frac{1}{2}E(R)$

and $\text{Var}\left(\frac{1}{2}R\right) = \frac{1}{4}\text{Var}(R)$ often seen. However, some candidates ignored the 'hence' and used the distribution for $\frac{1}{2}R$ together with methods already employed in part (b)(i) in order to find the answers.

The vast majority of candidates knew what was expected of them in part (c) and many fully correct answers were seen.

Q18.

Whilst full marks for answers to this type of question were more common than on previous papers, the majority of candidates scored less than half marks with the weaker candidates often scoring at most one mark. In part (a), most candidates found the correct value for the mean though, sadly, $5(2R - 1) = 10R - 1$ was not uncommon. In attempting to find the variance, candidates used $5(2\text{Var}(R) - 1)$ or multipliers for $\text{Var}(R)$ of $(5^2 \times 2)$ or (5×2^2) rather than 10^2 . A minority of candidates attempted to work from first principles but rarely with much success. Many candidates simply omitted part (b)(i) whilst others wrote down

values of $\frac{60}{P(R=r)}$. Those candidates who wrote down the correct probability distribution, then scored the mark in part (b)(ii). Surprisingly, a small number of candidates recovered the work in part (b)(i) in answering part (b)(ii). Earlier problems within the question rarely deterred candidates from attempting, often successfully, the final part, though not squaring 24.2 before subtraction from 759.4 was not unusual.

Q19.

In answering part (a), most candidates read 'explain' as 'show' and calculated $\frac{\sum d_i}{9}$ or $\frac{\sum d_i p_i}{1}$ with all $p_1 = \frac{1}{9}$ due to random selection. Such answers usually scored the two marks available. Answers to part (b) were generally correct with most candidates offering

the answer $\frac{20}{3}$ or $6\frac{2}{3}$ though a minority of candidates subtracted 5 rather than 5^2 . Answers to part (c) often failed to score marks through incorrect algebra, for example $3(E(D) + 5) = 3E(D) + 5$; or through incorrect application of the Var operator, for example $\text{Var}(N) = 3\text{Var}(D)$ or $3\text{Var}(D) + 5$ or even $3E(D) + 5$.

Q20.

Most candidates only scored some marks in part (a). Even here, $5 \times 3 - 1 = 14$ was not uncommon for the mean, as was $5(1 - 1) = 0$ for the variance. Valid attempts at part (b)

were extremely rare, with most candidates who made an attempt resorting to $\frac{12}{3} = 4$ for

the mean and either $\frac{12}{1}$ or $\frac{144}{1}$ for the variance. Where correct answers were seen, they were usually as a result of simply tabulating the probability distribution for R^{-1} or $12R^{-1}$ and then using the formulae for mean and variance.

Q21.

This question proved to be the most difficult on the paper for many candidates with few achieving full marks. In part (a), some candidates felt it to be unnecessary to simplify $[4T + (2T + 10)]$ or even $[2T + 2T + (T + 5) + (T + 5)]$. As a result, such candidates were able to find correctly that $E(P) = 40$ but failed to find the correct value for $\text{Var}(P)$. A minority of candidates even appeared to be unaware of the formula for the perimeter of a rectangle.

When attempted answers to part (b) (i) were seen, very few candidates appeared to be aware that $\text{Var}(T) = E(T^2) - (E(T))^2$ and so opted for $E(T^2) = 2 \times (E(T))^2$ to obtain the given answer of 50. In part (b) (ii), as in part (a), many candidates left the area unsimplified as $[2T \times (T + 5)]$ and then calculated $E(A)$ as $2 \times E(T) \times (E(T) + 5) = 2 \times 5 \times 10 = 100$.



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